Course Reminders

Due Sunday, Nov 1 (11:59 PM)

- A4
- Q5
- Mid-course survey (link also on Canvas; EC)
- Weekly Project Form (EC)

Notes:

- Inference II material shows up in *A5* (due *next* Sun, Nov 8)
- A2 scores posted (Canvas); fetch feedback in datahub

Inference II:

Relationships Between Variables

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Approaches to Inference

CORRELATION

ASSOCIATION BETWEEN VARIABLES

i.e. Pearson Correlation, Spearman Correlation, chi-square test

COMPARISON OF MEANS

DIFFERENCE IN MEANS BETWEEN VARIABLES

i.e. t-test, ANOVA

REGRESSION

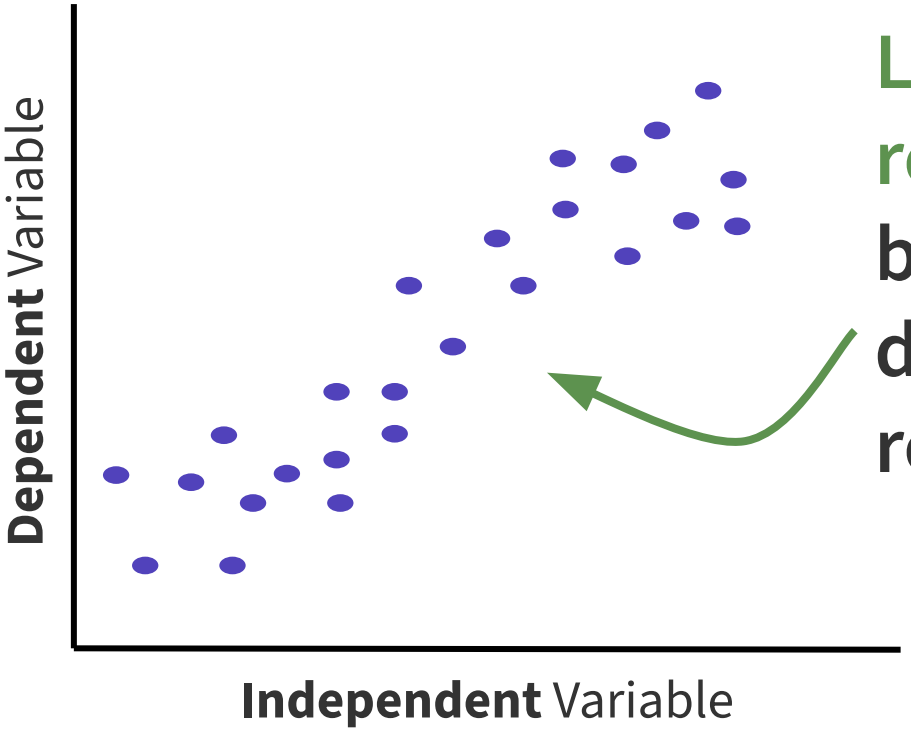
DOES CHANGE IN ONE VARIABLE MEAN CHANGE IN ANOTHER?

I.e. simple regression, multiple regression

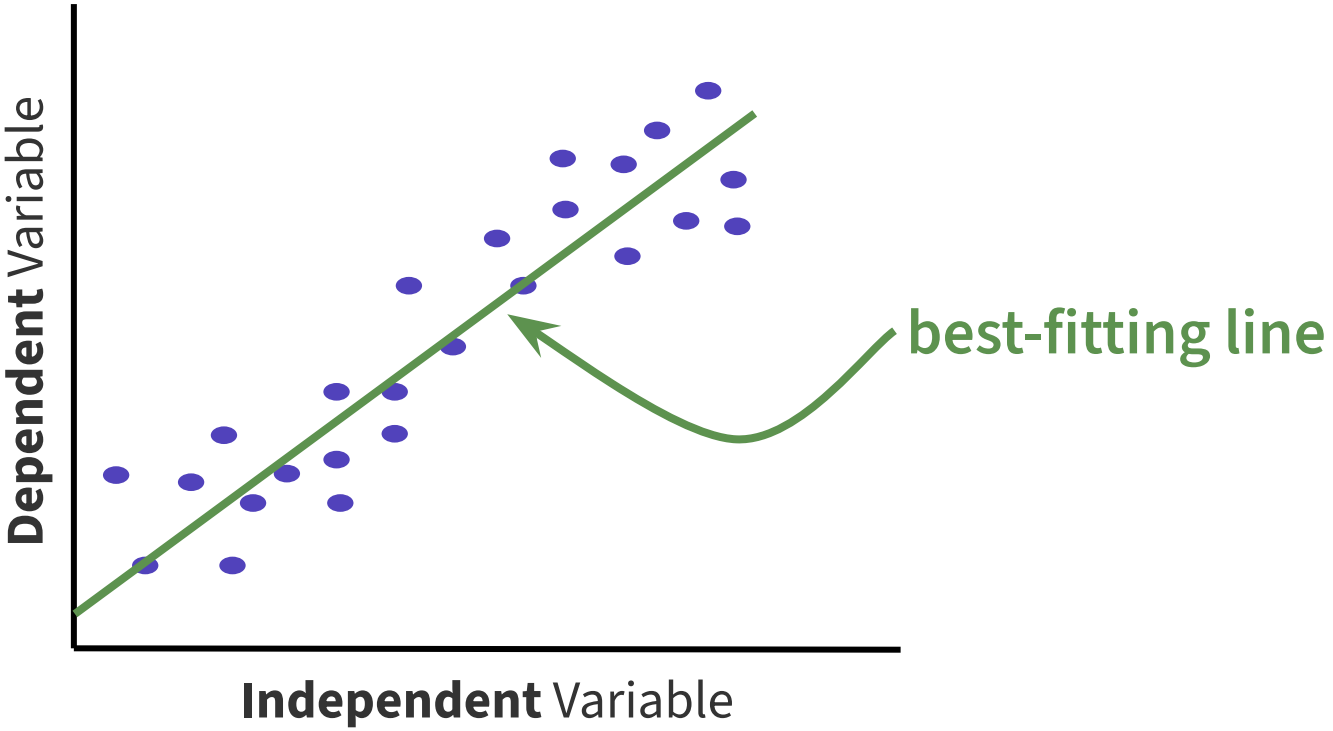
NON-PARAMETRIC TESTS

FOR WHEN ASSUMPTIONS IN THESE OTHER 3 CATEGORIES ARE NOT MET

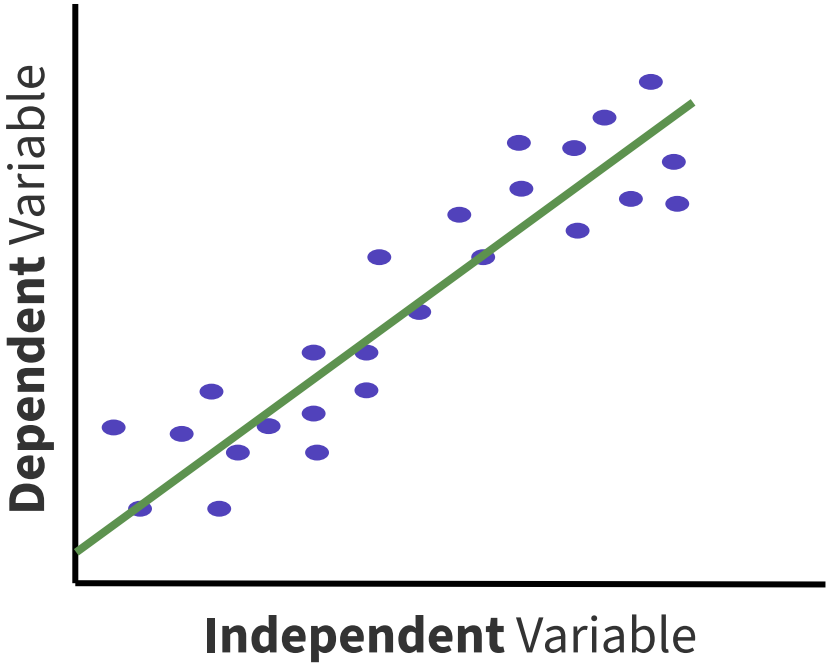
i.e. Wilcoxon rank-sum test, Wilcoxon sign-rank test, sign test



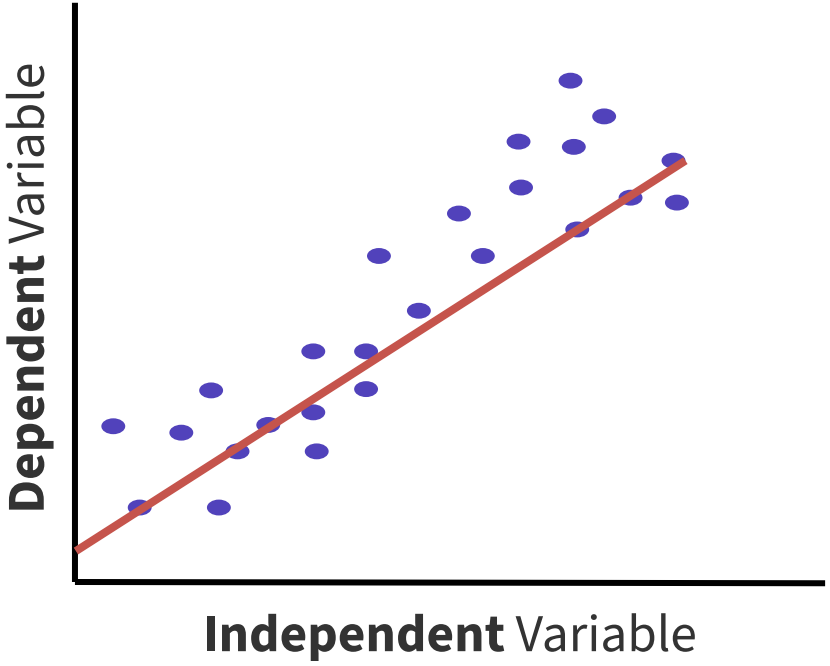
Linear regression can be used to describe this relationship

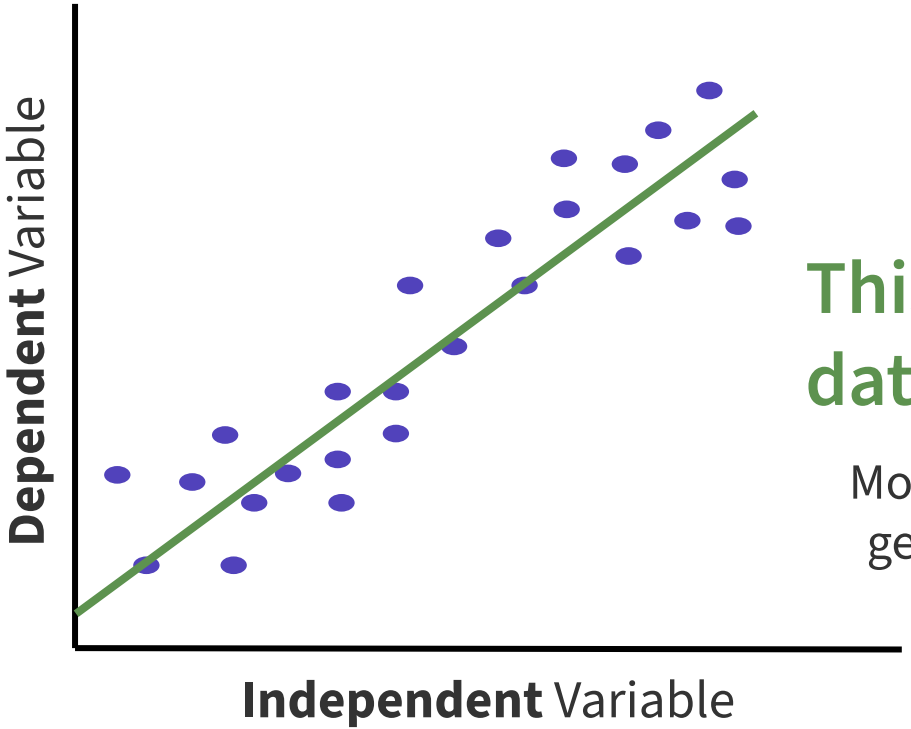

Dependent Variable
Independent Variable
best-fitting line

Best-fitting line



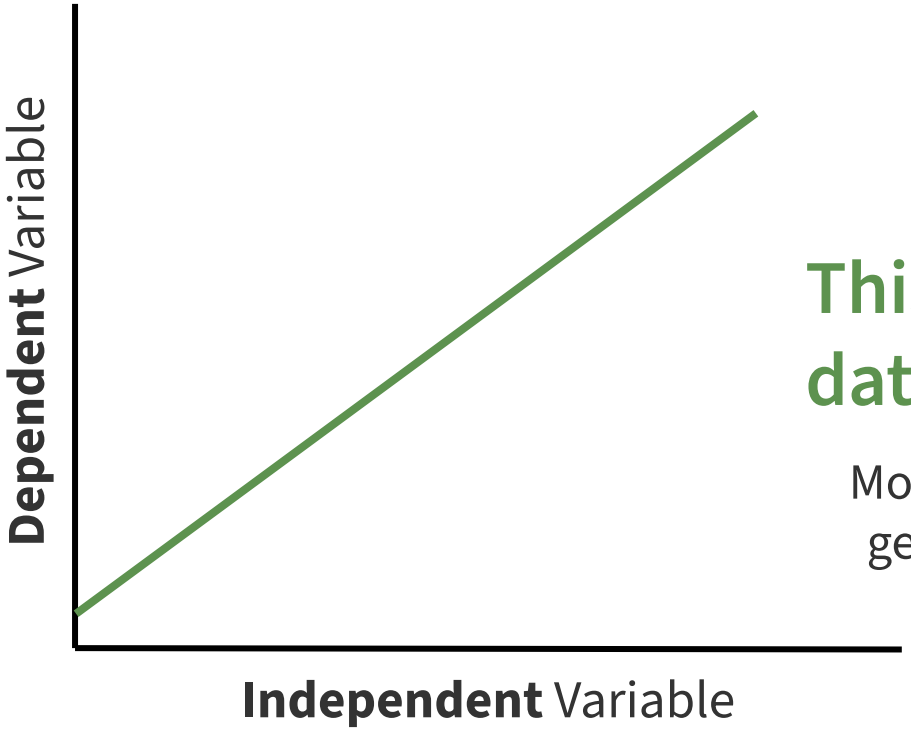
NOT a best-fitting line





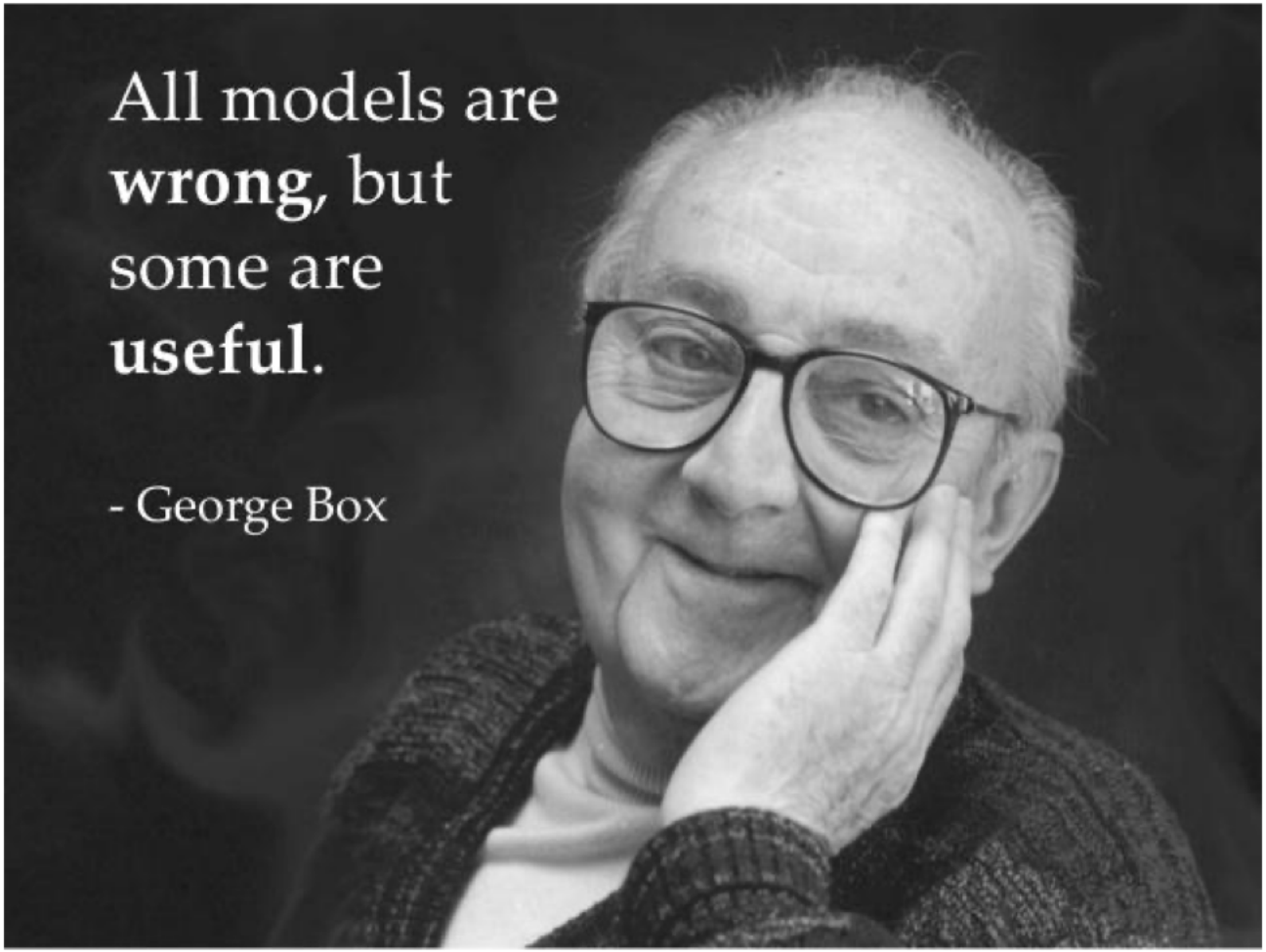
This line is a model of the data

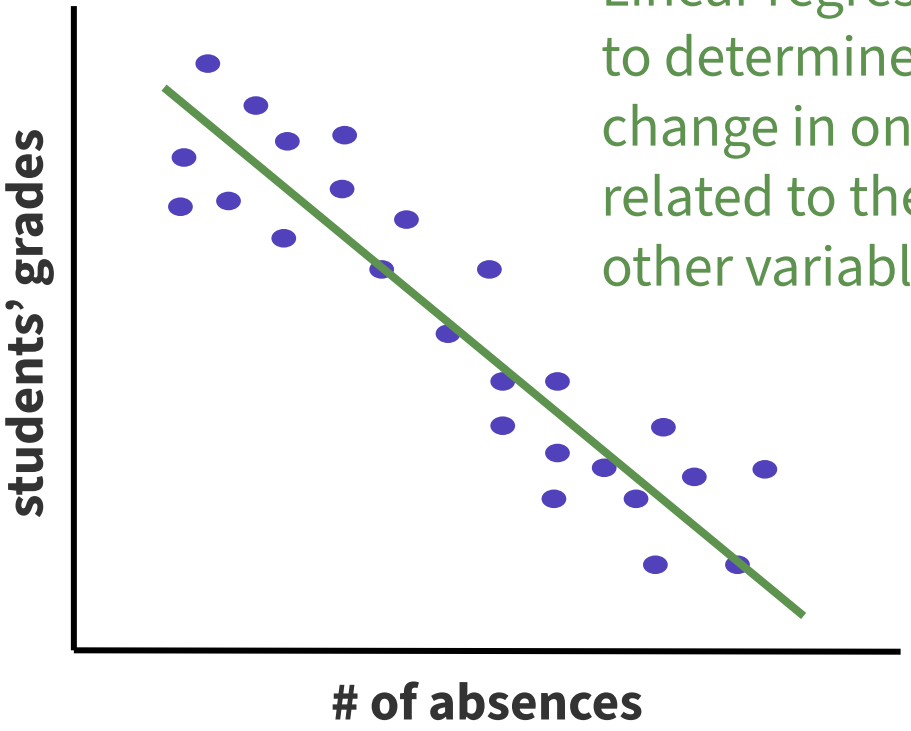
Models are mathematical equations generated to *represent* the real life situation



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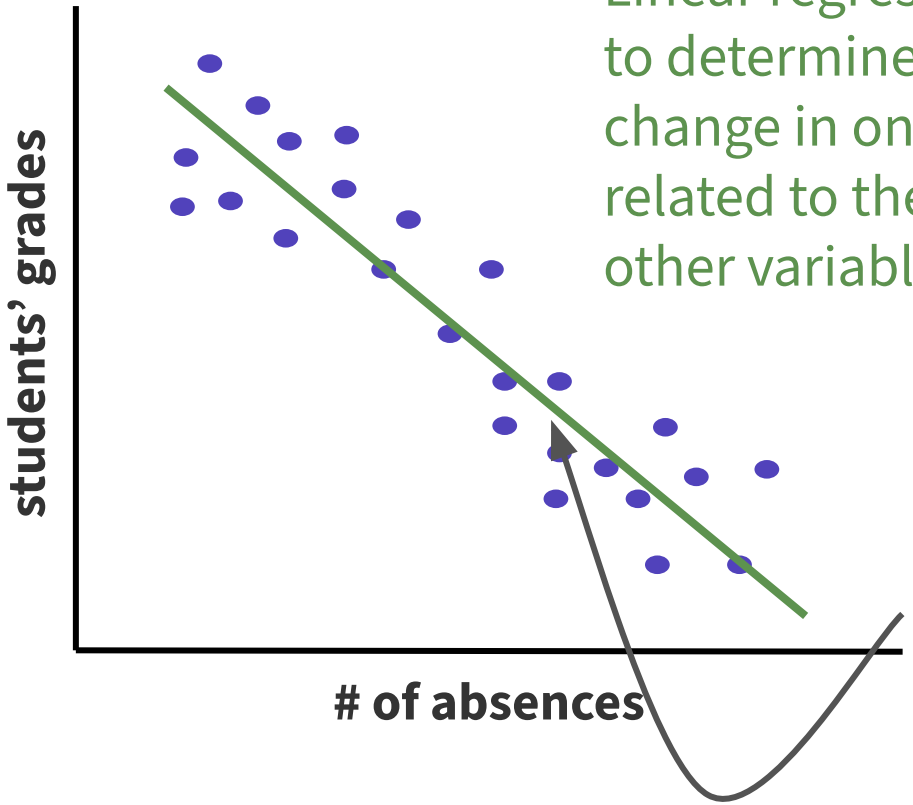
Models are mathematical equations generated to *represent* the real life situation


All models are
wrong, but
some are
useful.
- George Box



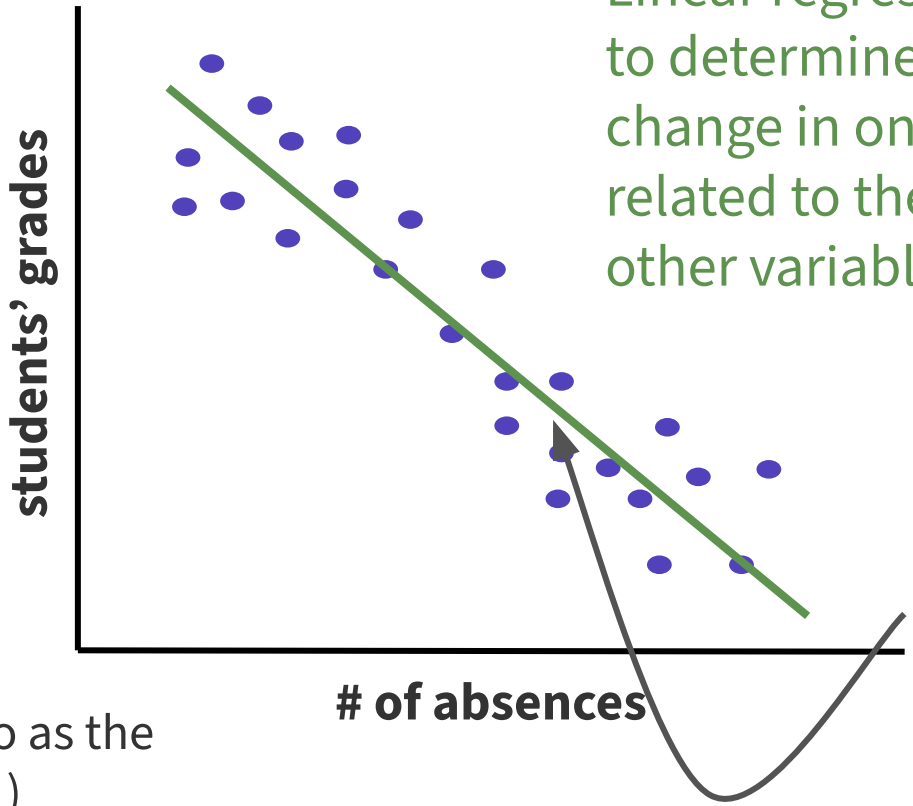
Linear regression can be used to determine whether a change in one variable is related to the change in the other variable

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The magnitude of the relationship is measured by the slope of the line

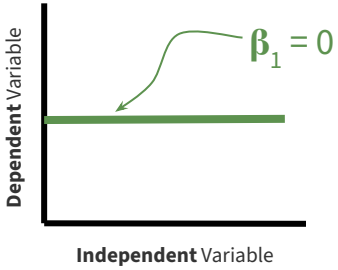
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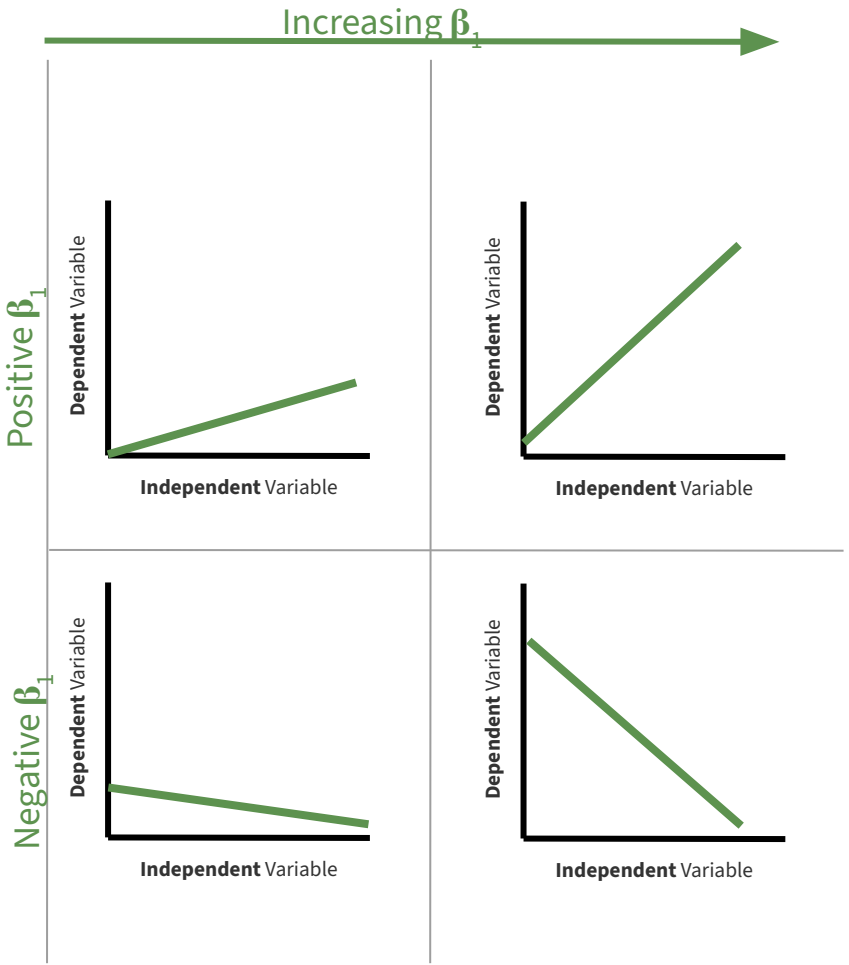


The magnitude of the relationship is measured by the slope of the line

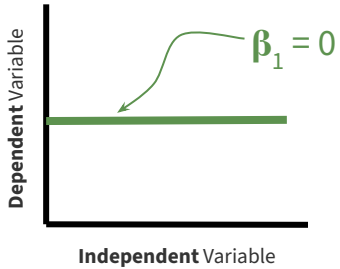
This is also referred to as the model's effect size (β_1)

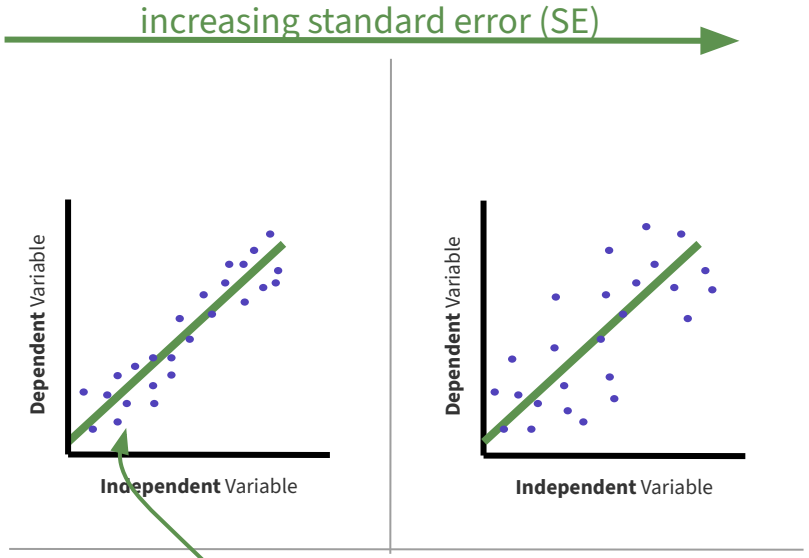
Effect size (β_1) can be estimated using the slope of the line





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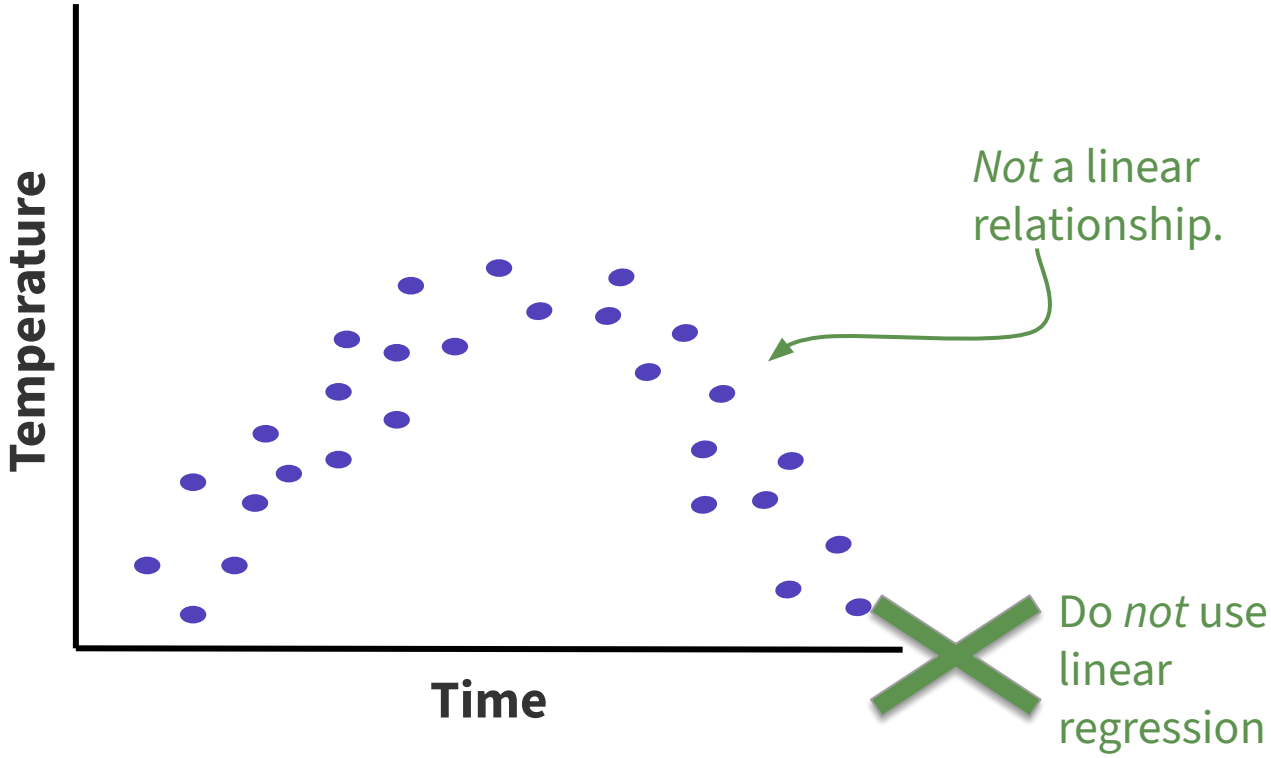




The *closer* the points are to the regression line, the *less uncertain* we are in our estimate

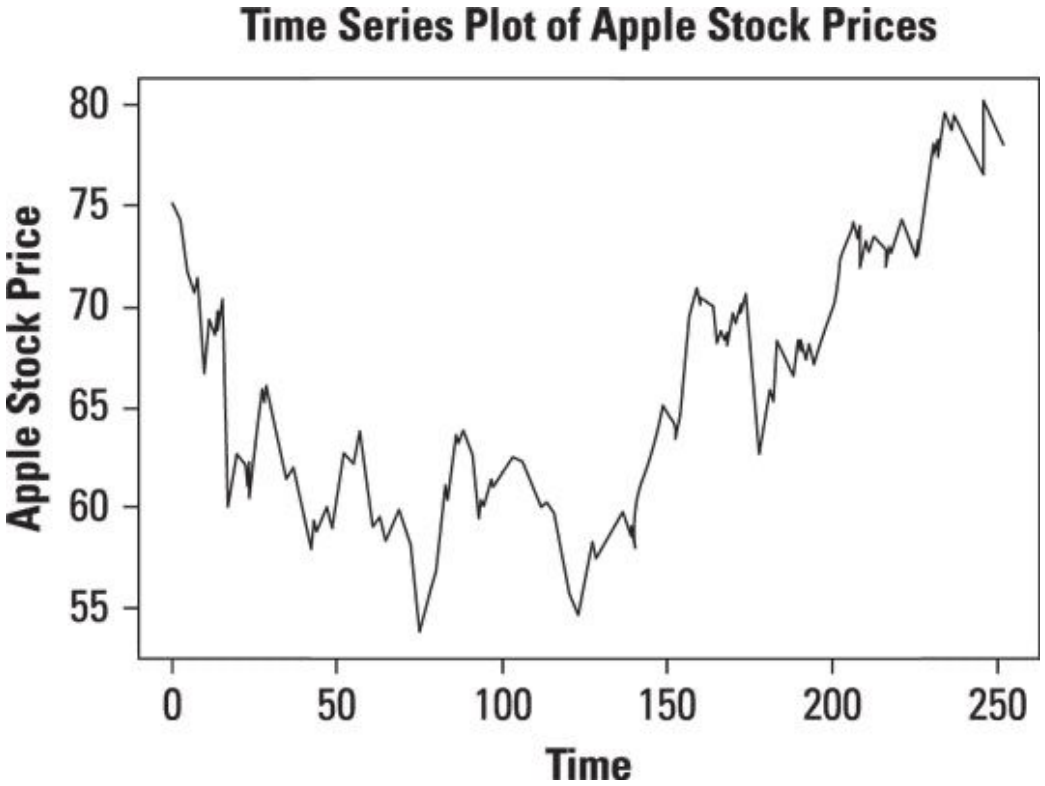
Assumptions of linear regression

1. Linear relationship
2. No multicollinearity
3. No auto-correlation
4. Homoscedasticity

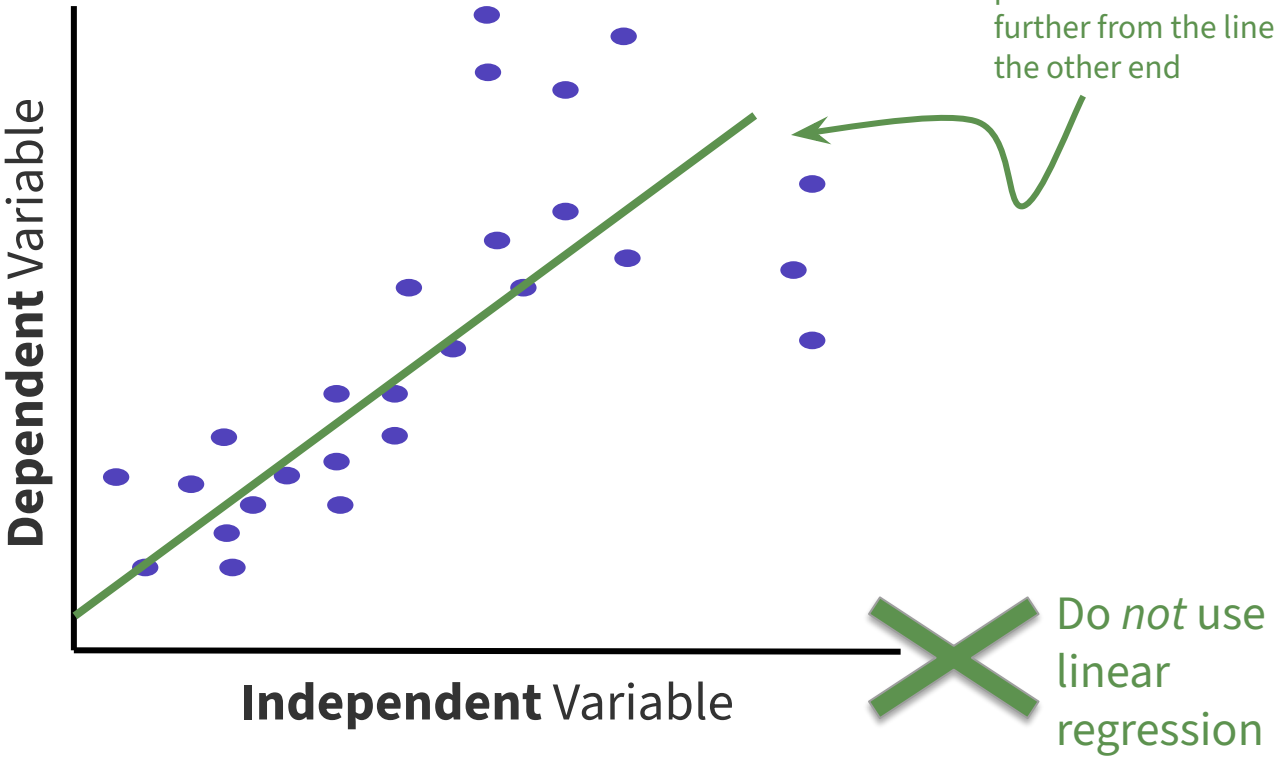

Temperature
Time
Not a linear relationship.
Do *not* use linear regression

Linear regression assumes no multicollinearity. Multicollinearity occurs when the independent variables (in multiple linear regression) are too highly correlated with each other.

Autocorrelation occurs when the observations are *not* independent of one another (i.e. stock prices)



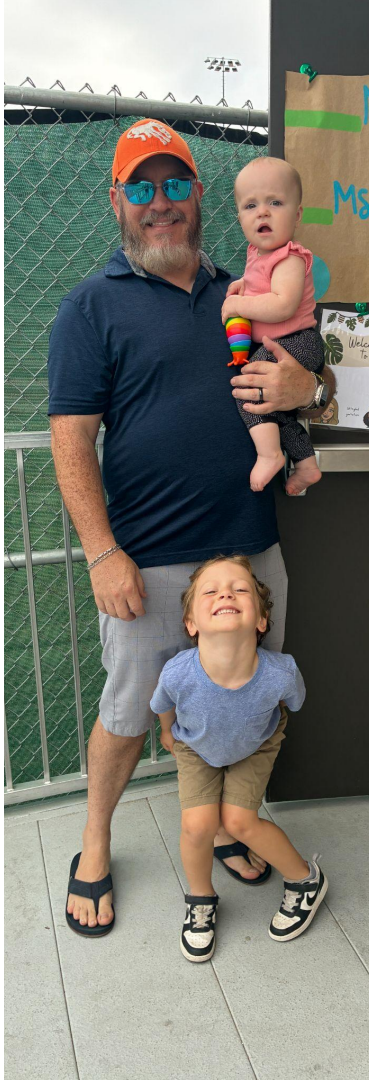
https://www.dummies.com/programming/big-data/data-science/autocorrelation-plots-graphical-technique-for-statistical-data//


Not homoscedastic:
points at this end are much further from the line than at the other end
Dependent Variable
Independent Variable
Do *not* use linear regression

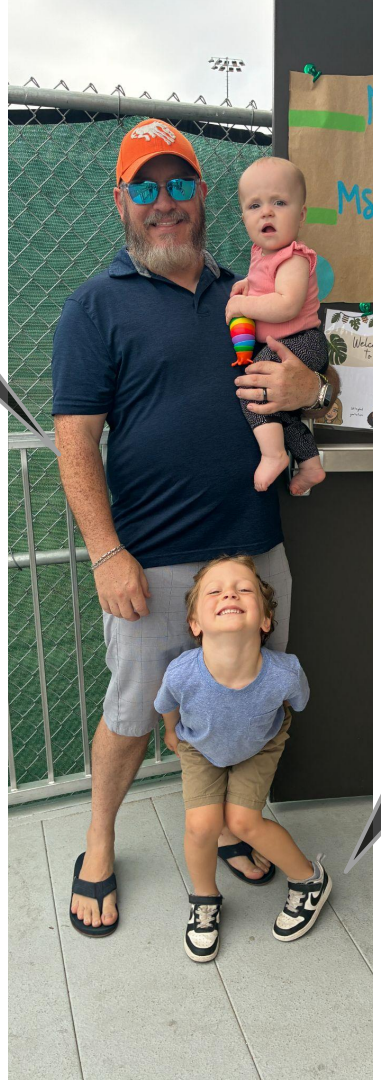
Takes into account the effect size (β_1) and the SE

p-value : the probability of getting the observed results (or results more extreme) by chance alone

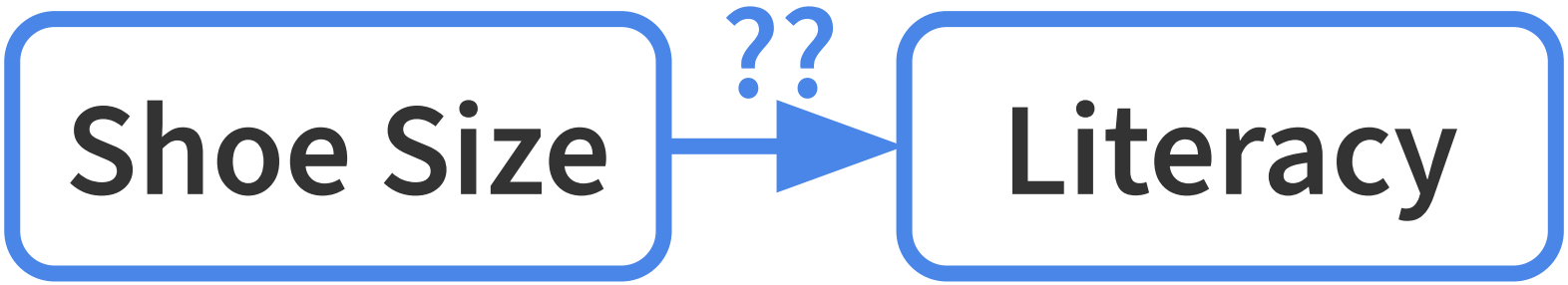
Confounding



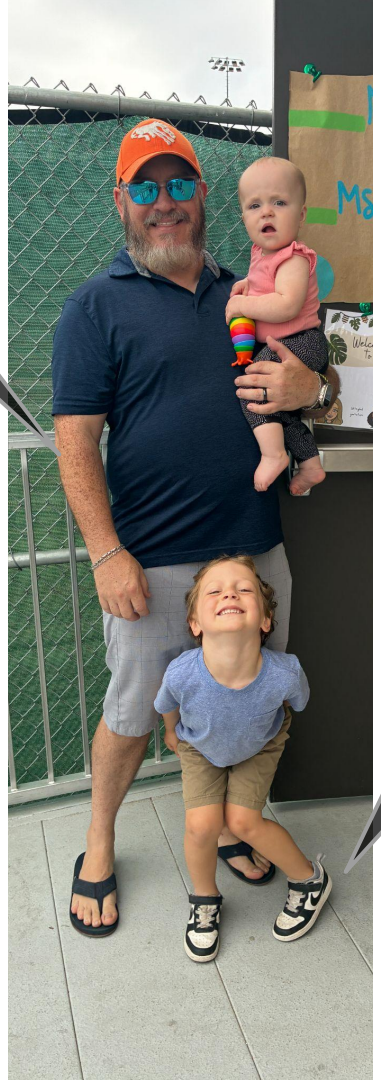
Big shoes
Literate



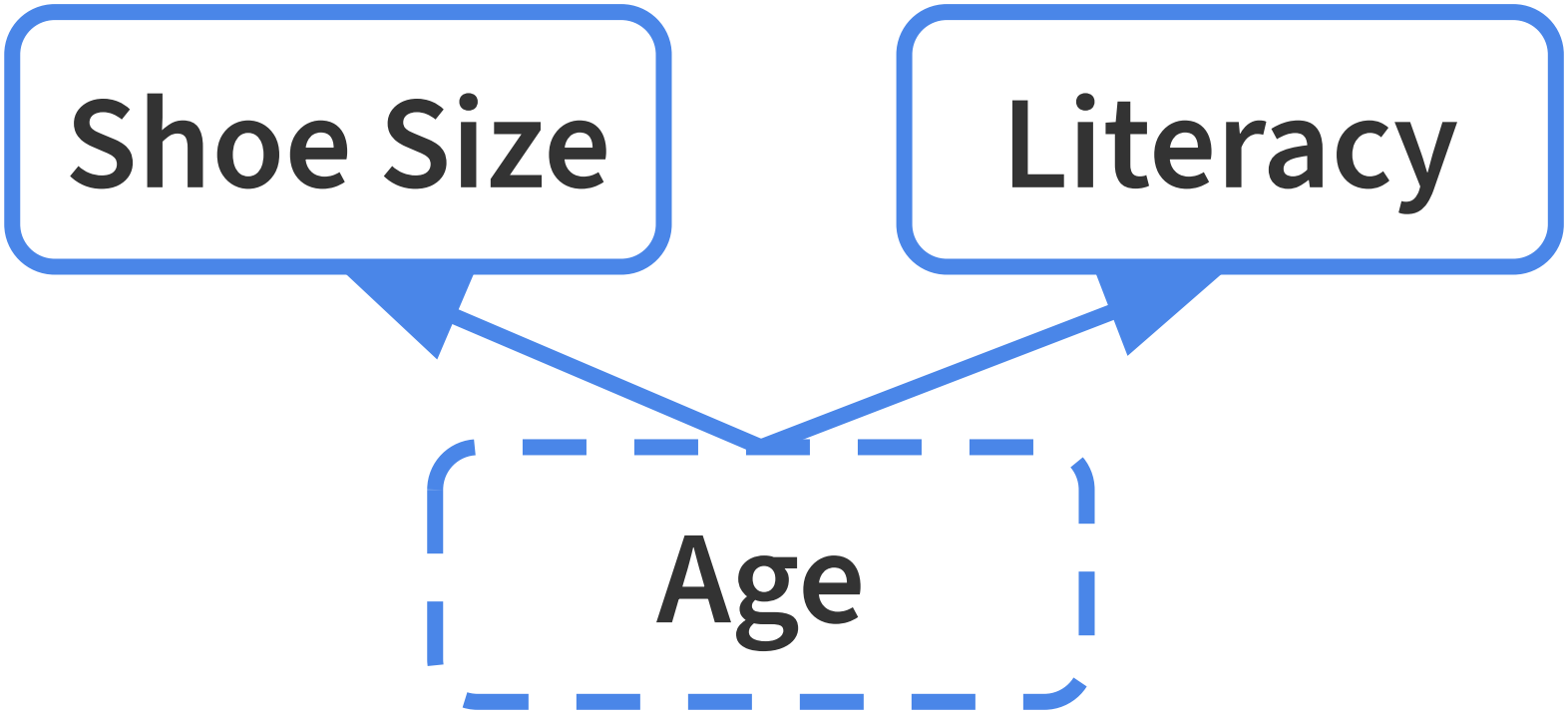
Small shoes
Not literate

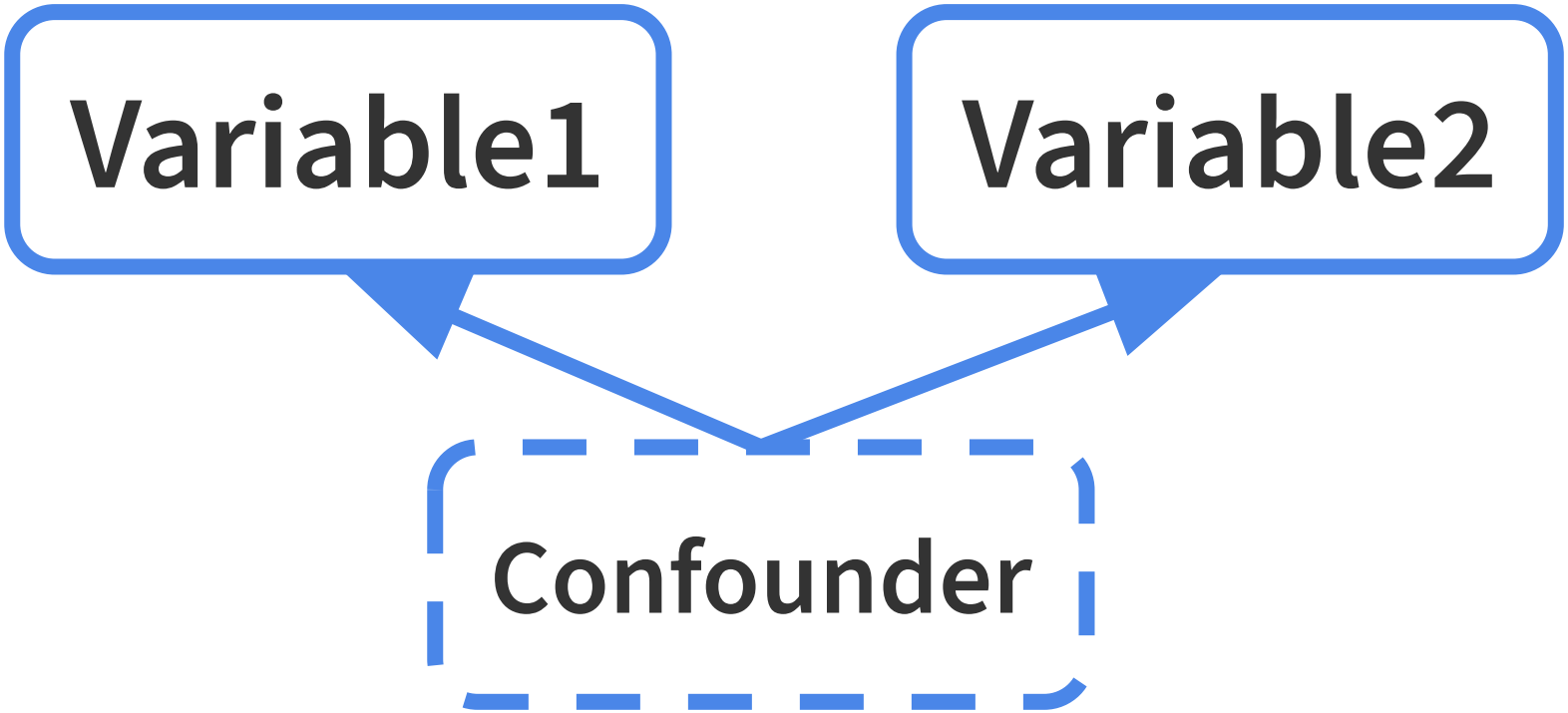

Shoe Size
??
Literacy

Big shoes
Literate
Adult



Small shoes
Not literate
Child


Shoe Size
Literacy
Age


Variable1
Variable2
Confounder

https://forms.gle/wd8jQvVNzMb9o99X9

Confounding

popsicles → crime rate





Your analysis sees an increase in crime rate whenever popsicle sales increase. What could confound this analysis?

A
popsicle preference

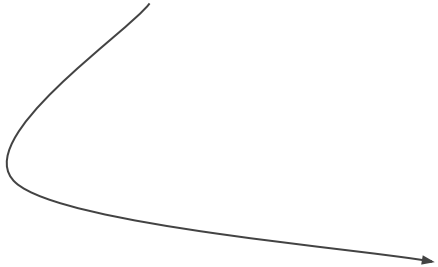
B
new gun laws

C
temperature

D
changes in popsicle prices

E
new law enforcement officers

Multiple regression is one way to account for confounders. We'll try it in the notebook.



Ignoring confounders will lead you to draw incorrect conclusions from your analyses